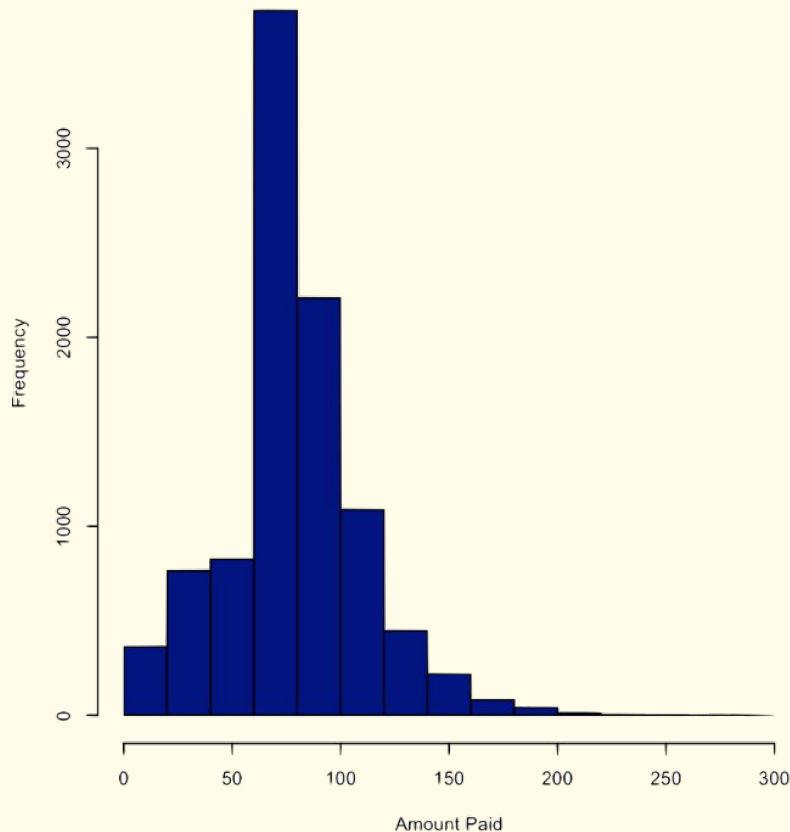

MISO Case

Group 20.7: Ina Lin, Cindy Chou, Khwaish Gohil, Yin Zhi & Priya Thakore

Mar 3

Descriptive Statistics: Video



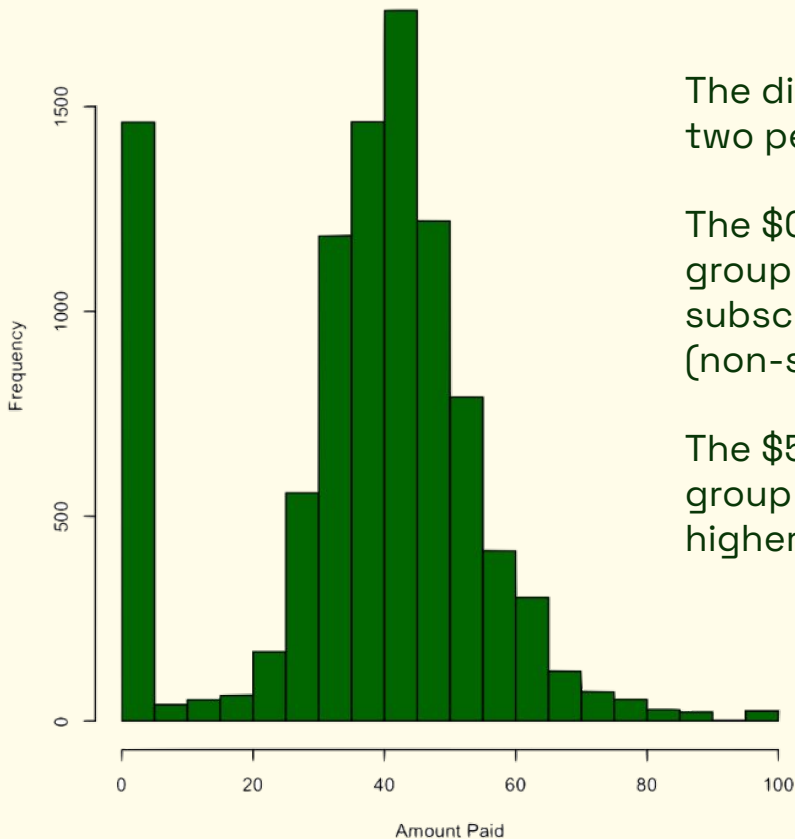
The distribution is right-skewed, with most customers paying between \$40 and \$100.

There are a few outliers where customers are paying over \$200

This suggests that some customers opt for premium video services

Higher mean and wide range suggest significant variation among customers

Descriptive Statistics: Internet



The distribution is bimodal, with two peaks around \$0 and \$50

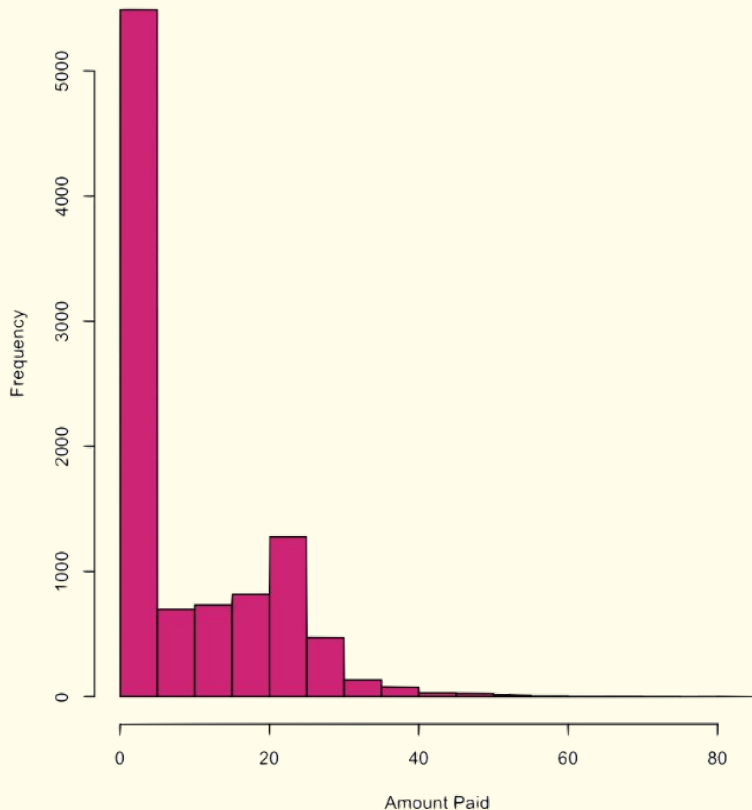
The \$0 peak represents a large group of users who do not subscribe to internet service (non-subscribers)

The \$50 peak suggests a large group of customers opting for higher plan (active subscribers)

A long tail distribution suggests the presence of premium users paying over \$80, who may form a high-value segment

These patterns highlight the need to differentiate clusters based on internet usage levels for more precise targeting

Descriptive Statistics: Phone



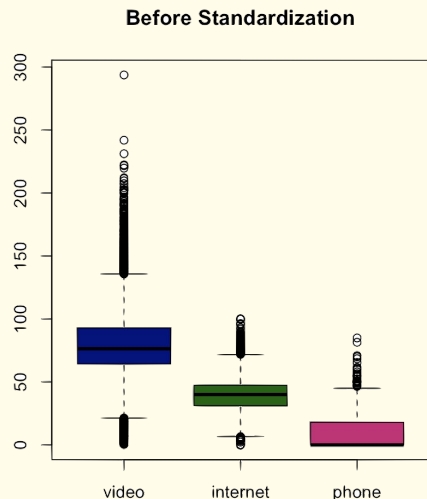
Majority of households pay \$0 for phone service, indicating a large segment of non users who have likely stopped using landlines

Among paying customers, most spend between \$10-\$30, with few opting for premium plans above \$30

Clustering should separate non-subscribers from active users, to refine marketing strategies based on phone usage levels

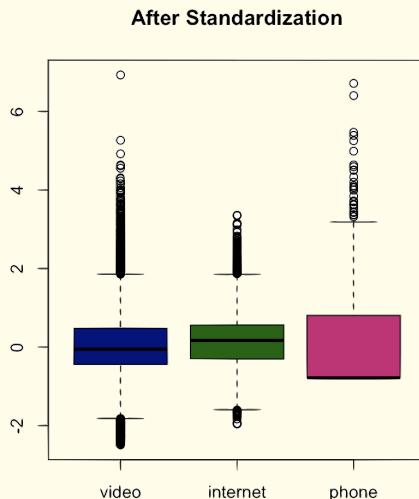
Standardization

We should **not standardize** video, internet and phone usage because it would eliminate the impact of scale differences, which are crucial for our analysis



The variation in video payments dominates due to its higher SD (31.14), making it a primary differentiation

Variable	SD
Video	31.14
Internet	18.85
Phone	11.33



All payment types contribute equally to clustering, potentially overlooking important natural differences

Variable	SD
Video	1.00
Internet	1.00
Phone	1.00

Preserving Meaningful Variations

- Without standardization, the natural differences in usage amounts reflect real world spending patterns
- Video payments higher SD might indicate greater variation in spending

Ensuring Proper Weighting

- If standardization is applied, all variables will have equal influence, potentially hiding important distinctions

Avoiding Misrepresentation

- Standardizing could distort the relative importance of each category
- For example, video payments naturally have higher variability, which could be a key insight for clustering

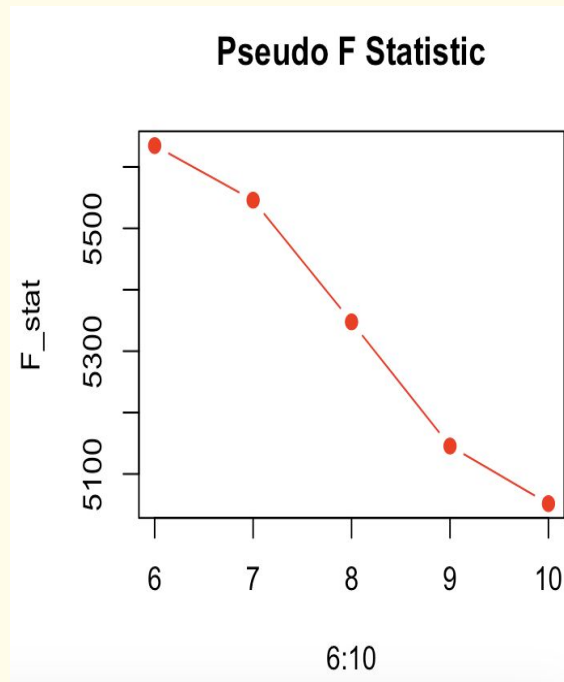
Cluster Solutions Using K-Means

We suggest that 6 clusters may be the best way to segment. As can be seen from the graph, 6 clusters have the highest F value compared to other clusters. The relevant high SSB and low SSE also indicate that 6 clusters fit the data we have.

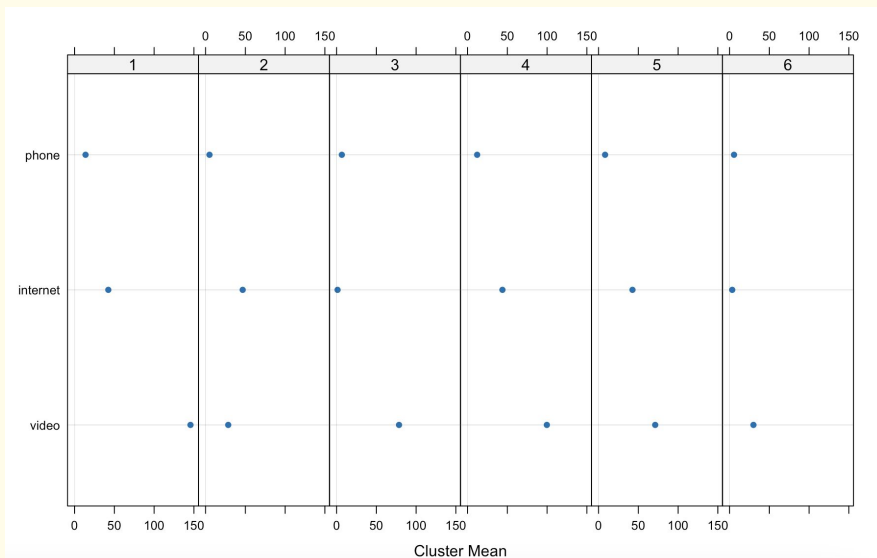
It can also be seen from the R square value that it accounts for 74% of the situation, which is also the highest among all the other clusters.

	n	Pct	video	internet	phone	RMSE
1	717	0.07	145.70	42.40	13.79	17.7173
2	1108	0.11	28.47	46.65	4.93	11.5089
3	1197	0.12	78.53	1.24	6.63	10.3423
4	2344	0.24	99.68	43.86	12.07	11.2455
5	4036	0.41	71.22	42.66	8.22	9.7708
6	378	0.04	30.05	3.39	5.64	10.2421
	9780	1.00	77.96	36.79	8.88	11.1740

SSE= 3661119 ; SSB= 10553167 ; SST= 14214286
R-Squared = 0.7424339
Pseudo F = 5634.707



Cluster Profiles



1. Highest in video, slightly low in Internet, low in phone(although highest among all)
2. Lowest in video, sightly high in Internet, low in phone(although lowest among all)
3. Above average in video, lowest in Internet but also on the average level, slightly above average in phone
4. Second highest in video, above average in Internet, high in phone
5. Slightly lower than average in video, slight higher than average in internet
6. Second lowest in video, very low in internet (second lowest)

cluster	avg_age	avg_income	avg_size	married_pct	avg_children	children_pct
<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	53.4	5.87	3.26	67.9	0.392
2	2	46.2	4.31	2.49	44.1	0.364
3	3	62.3	3.83	2.54	49.1	0.243
4	4	52.1	5.24	3.02	59.2	0.396
5	5	50.1	4.48	2.64	49.1	0.341
6	6	57.8	3.67	2.29	44.7	0.220

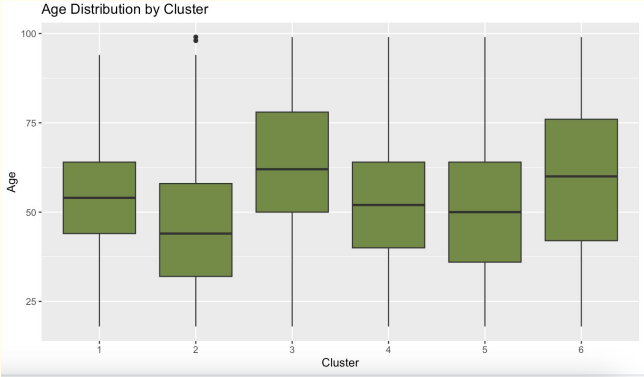
Cluster Profiles: Demographics

married_pct	avg_children	n
<dbl>	<dbl>	<int>
67.9	0.392	717
44.1	0.364	1108
49.1	0.243	1197
59.2	0.396	2344
49.1	0.341	4036
44.7	0.220	378

	cluster	avg_age	avg_income	avg_size	married_pct	avg_children	children_pct
	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	53.4	5.87	3.26	67.9	0.392	39.2
2	2	46.2	4.31	2.49	44.1	0.364	36.4
3	3	62.3	3.83	2.54	49.1	0.243	24.3
4	4	52.1	5.24	3.02	59.2	0.396	39.6
5	5	50.1	4.48	2.64	49.1	0.341	34.1
6	6	57.8	3.67	2.29	44.7	0.220	22.0

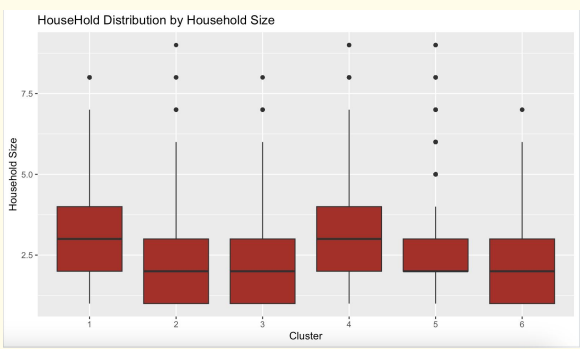
Cluster Profiles: Demographics

Age



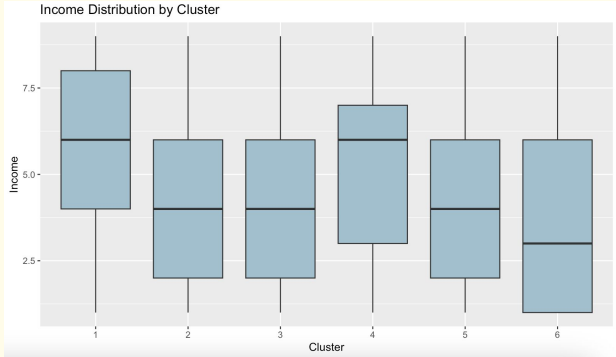
Cluster	Average Age
1	53.4
2	46.2
3	62.3
4	52.1
5	50.1
6	57.8

Household Size



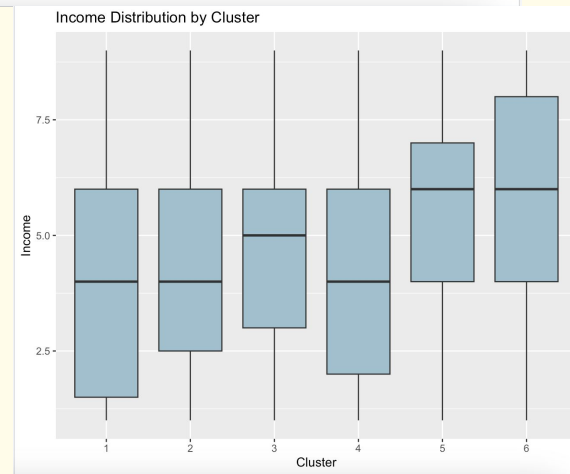
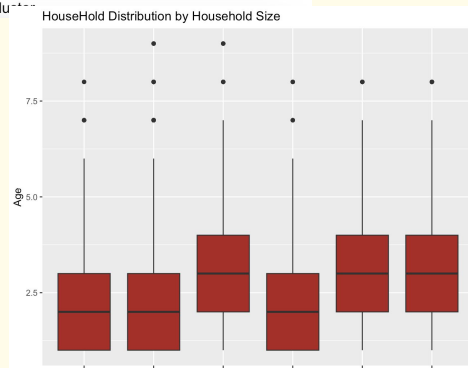
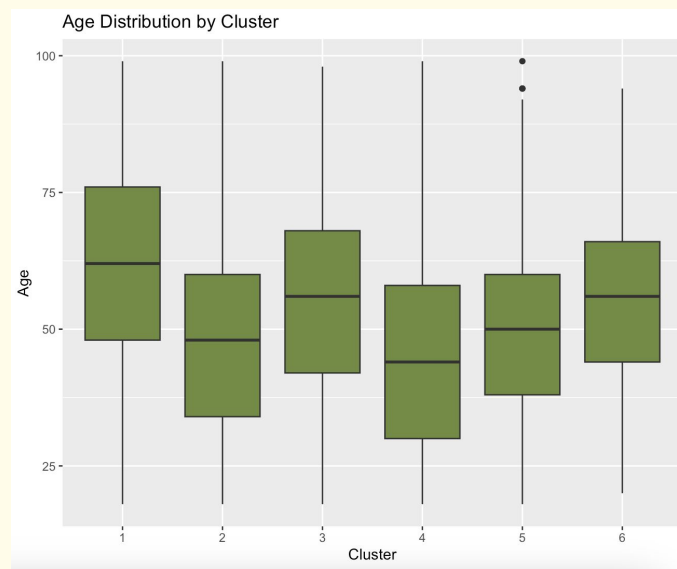
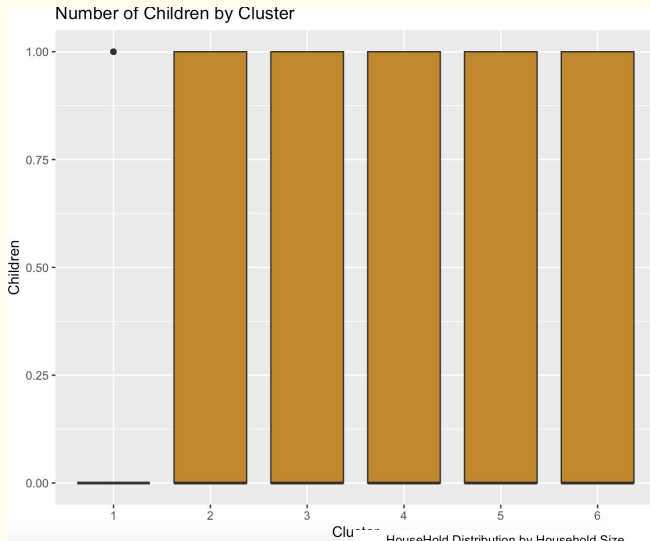
Cluster	Average Household Size
1	3.26
2	2.49
3	2.54
4	3.02
5	2.64
6	2.29

Income



Cluster	Average Income Group
1	5.87
2	4.31
3	3.83
4	5.24
5	4.48
6	3.67

Cluster Profiles



Cluster Profiles

Profile the clusters from your best cluster solution on the demographic variables to get a better understanding of the clusters. Where are there large differences? Can you get a sense of the typical person in each cluster?

Cluster 1: Premium Users

Cluster 2: Video Haters

Cluster 3: Retired

Cluster 4: Established Communicators (High Income)

Cluster 5: Well-rounded communicators

Cluster 6: Light Users

Cluster 1: Premium Users

- Demographics
 - Average age 53
 - Nearly 70% married, with a bigger household(with children)
 - Highest income
- Usage
 - Highest in video, slightly low in Internet, low in phone(although highest among all)
- Possible characteristics:
 - Have more disposable income to spend on MSO services
 - Plays video all day as background sounds for entertainment
 - Use internet and calling through cell phone for business
- Possible offers:
 - “Unlimited phone, video, internet bundle package” that charges the highest price

Cluster 2: Video Haters

- Demographics
 - Average age 46 (youngest)
 - Around 44% married, small household
 - An average income across clusters
- Usage
 - Lowest in video, highest in Internet, low in phone(lowest among all)
- Possible characteristics:
 - Prefer form of entertainment/news source through internet than video
 - Enjoys and uses internet, phone is just there if needed
- Possible offers:
 - Internet and phone bundle package +basic video channels

Cluster 3: Retired

- Demographics
 - Average age of 62.3, the highest among all the clusters
 - Below average income compared to other clusters
- Usage
 - Largely above average in video, about average in Internet but the lowest among all the clusters
 - slightly above average in phone
- Possible characteristics:
 - Soon to be retired
 - Empty nesters
 - More used to use videos, but also trying phones and Internet
- Possible offers:
 - "Stay Connected" Plan: Special rates for retirees or empty nesters, offering a cost-effective plan to match below-average income.

Cluster 4: Established Communicators (High Income)

- Demographics
 - Average age slightly about 52.1
 - About 59.2% married, with 0-1 child
 - Above average income, also high
- Usage
 - About average for all
 - Higher than average in video (second highest) and high in landline phone
- Possible characteristics:
 - Families that have different communication needs
- Possible offers:
 - Multi-Device Bundles: packages that include home internet, mobile and video services, ensuring seamless connectivity for the whole family

Cluster 5: Well-Rounded Communicators

- Demographics
 - Average age around 50
 - Nearly 50% married, with 0-1 child
 - Ranks third in income group
- Usage
 - Slightly lower than average in video
 - Slight higher than average in internet
- Possible characteristics
 - Families that have different communication needs
- Possible offers
 - “Family plan”: phone, internet, video bundle with additional marketing efforts on selling them a higher-tiered internet service

Cluster 6: Light Users

- Demographics
 - Avg age 57.8, mostly no children
 - Lower income group
 - Near 45% married
- Usage
 - Significantly lower than average across all devices
- Possible characteristics
 - Minimal needs for internet and video
 - Phone is there when it's needed
- Possible offers
 - Basic Essentials Package – A low-cost bundle with minimal internet, basic TV channels, and a landline.
 - Streaming Starter Pack – A small, curated selection of streaming options at a lower price to introduce them to digital services.

Cluster 7: Video Haters

- Demographics
 - Avg 1 children, one-three person household
 - Avg income
 - Avg age (45-55)
- Usage
 - Very low video
 - Above average for internet
 - Slightly below average phone
- Possible characteristics:
 - Prefer other form of entertainment/news source than video
 - Enjoys and uses internet, phone is just there if needed
- Possible offers:
 - Internet and phone bundle package +basic video channels

Additional Variables

for Personalization in Cluster Analysis

Variable 1: Monthly Data Usage

Reason: We can combine this aspect with different clusters' preferences. For example, cluster 7 has a higher preference on Internet compared with phone and video, although not high. If they also have higher data usage (e.g. more GBs downloaded), it may be a good idea to offer them Internet offers. However, if their data usage is low, then Internet offers may not be strongly effective.

Variable 2: Subscription length

Reason: Help to establish personalized retention and loyalty strategies. Knowing subscription length can also provide insights into how long customers typically stay within each cluster, allowing the MSO to predict the future behavior of customers, anticipating potential churn risks. For example: cluster 6 are premium users, inclined towards high usage of video, offering them exclusive benefits such as early access to new features and Loyalty rewards to these potential long-term customers. Similarly, for clusters like 2, 3, 4 and 5 who are a bit less video-focused, retention incentives like offering free services or a trial of premium features will encourage them to stick continue and boost customer loyalty.

Additional Variables

Variable 3: Amount of Connected Devices

Reason: Users who connected with multiple devices and who connected with single device may have different preferences. For example, cluster 1 now shows high preference in videos while only the average level in other two devices, then it may be better to offer them with video offers. However, if we have the amount of connected devices, we may find that these people also connected with internet, then an additional Internet offer may also be effective. By integrating both device-specific engagement data and the overall number of connected devices, we can develop more precise marketing strategies, ensuring that users receive offers that align with their actual usage patterns and preferences.